This guide is for anyone wanting to complete mathematics from the beginning to major fields of mathematics that will give you the tools to read the current research papers in mathematics. You can start from anywhere you want. You may get the book by any means – no need to waste money.

We need to start with foundational mathematics. The best book I can recommend is *Prealgebra* by Rusczyk. The best book for the topic period. It doesn't hold your hand and also shows you some of the fun tricks in math. Even if you don't like mathematics everyone should be exposed to easiest level of competition mathematics.

If you wish to do competition mathematics or in high school with time to spare or equivalent you must continue the AOPS series as it is the best series of texts available for someone wanting practice as well as face challenges in basic mathematics. You may skip the below books until I reach calculus portion.

If not then I must recommend *Basic Mathematics* by Serge Lang. While it is a precalculus book, it completely encapsulates all of high school mathematics in one book and is an exceptional first pass over most topics.

After this go through *Algebra, Functions and Graphs, Method of Coordinates, Trigonometry* and *Geometry* from the Gelfand Correspondence Program and use appropriate problem books for each topic covered. Good practice is key.

You may find some topics are missing from Gelfand series compared to AOPS series. Congratulations, you have discovered cross referencing. Keep doing this in the future. You may use all the books that have overlapping content as is the case above as you see fit. Instead of reading more books it may be better to pick up problem books at the precalculus and proof writing stage of your mathematics learning.

Please read *Book of Proof* by Hammack for introductory set theory and getting acclimated to proof writing which will be useful in abstract algebra and analysis.

You may question there is neither a discrete mathematics text nor any dedicated probability book that you might see in high school math exams. I believe you should just go through your school's materials and past exams for these topics because such topics are better covered at the university level. Only proof writing and precalculus should be emphasised at the high school level as it opens so many doors which the student himself/herself should find out. Curiosity is highest at this time after all.

This being a mathematics guide and all, I will be giving an introduction to analysis book instead of a standard university calculus text. But obviously 99% universities don't test analysis skills for majority of students in standard first year mathematics or other

degrees. For this reason, I think you should refer problem books like Schaum's Outlines or Stewart *Calculus* or Thomas *Calculus* or Liethold *Calculus 7* or *Calculus: A Complete Course* by Adams and Essex and equivalent books to get used to the computational side of calculus. There is no point in learning analysis if you can't actually compute integrals. This trick is called "going through computation before theory". You may use this trick for all incoming math topics like analysis and algebra whenever you face too high of a challenge. You might face this problem too. For example, say that I have recommended Author X's book on measure theory but Author X also has a book on introductory real analysis that has not been mentioned on this list. Then first go through the book I recommended on real analysis and then skim through Author X's book on introductory real analysis to get a feel for his style and finally attempt Author X's book on measure theory.

With that out of the way we can proceed with core undergraduate mathematics.

- Single variable analysis — *Calculus* by Spivak supplemented with *Understanding Analysis* by Abbott. You may want to read *How to Think About Analysis* by Alcock in your free time.
- Linear algebra — *Linear Algebra* by Friedberg, Insel and Spence supplemented with *Linear Algebra Done Right* by Axler.
- Abstract algebra — *Abstract Algebra* by Dummit and Foote or *Algebra* by Artin.
- Multivariable analysis — *Analysis on Manifolds* by Munkres and *Calculus on Manifolds* by Spivak supplemented with *Vector Calculus, Linear Algebra, and Differential Forms* by Hubbard and Hubbard. You may want to parse *Calculus* by Apostol in your free time.
- Undergraduate real analysis — *Principles of Real Analysis* by Rudin supplemented with *Real Mathematical Analysis* by Pugh. You should also read *Analysis* by Terence Tao.
- Undergraduate complex analysis — *Complex Analysis* by Stein and Sakarchi supplemented with *Complex Variables with Applications* by Brown and Churchill. Read *Visual Complex Analysis* by Needham for the big picture and the really beautiful applied complex analysis.
- Ordinary differential equations — *Ordinary Differential Equations and Dynamical Systems* by Teschl and *Ordinary Differential Equations* by Arnold.

The bridge to graduate analysis – *Real Analysis* by Stein and Sakarchi. The graduate core — for analysis use *Real and Complex Analysis* by Rudin, *Functions of One Complex Variable* by Conway and *Real Analysis: Modern Techniques and Their Applications* by Folland in that order. For algebra use either Algebra by Hungerford or Algebra by Lang referring to the other when stuck. You may use your own books at this stage if you were recommended any.

You can go into other branches of mathematics like number theory, analysis, probability and statistics, combinatorics, numerical analysis, differential equations and so on. You can approach them early on but if you are not completing a degree (i.e. self studying) it is better to divert after completing at least the undergraduate core first. I encourage you to make your own path from here onwards because the texts get increasingly more niche.

Obviously this looks like a reading list but a guide should pitch a study system. You should start with one book. Read a chapter and do its solved/unsolved examples by hand as well as the skipped proofs/skipped proof-steps by yourself. Try to engage in the direction/method presented in texts. If you find yourself pondering over a question for what seems hours on end, that means you are learning. In case you believe you are not learning mathematics and you may question why even study it if you wont use it ask yourself why breathe if you know you are going to die.

After reading this you may want to add in a few books maybe remove others or something else, I ask you to stop parsing roadmaps and start studying. The sooner the better. El Psy Congroo.